Arithmetic series

1 A sequence is a list of terms, and a series is the sum of the terms.

2

a $S_{15} = 450$ b $S_{15} = -45$ c $S_{15} = 105$ d $S_{15} = 465$

e $S_{15} = -660$ f $S_{15} = 615$

3 $S_{14} = \frac{273}{2}$

4

a $n = 18$ b $S_{18} = 450$

5 $S_n = n^2$

6

a $S_n = \frac{n}{2}(-5n + 13)$ b -954

7

a $S_{10} = 195$ b $S_{10} = 0$ c $S_{10} = 47.5$ d $S_{10} = -53.75$

e $S_{10} = 85$ f $S_{10} = 5$

8

a $d = -8$ b $S_{14} = -798$

9 $S_{20} = 210\pi$

10

a $S_{40} = 4920$ b $S_{40} = -2260$

11 $u_{20} = 62$

12 $n = 10$

13

a

i $d = 4$ ii $u_1 = 15$ iii $S_{25} = 1150$

b

i $d = \frac{2}{5}$ ii $u_1 = 15$ iii $S_{25} = 495$

c

i $d = \frac{1}{2}$ ii $u_1 = 15$ iii $S_{25} = 525$

d

i $d = 10$ ii $u_1 = -10$ iii $S_{25} = 2750$

e

i $u_1 = \frac{-7d+4}{2}$ and $u_1 = \frac{-13d+28}{2}$

ii $d = 4$

iii $u_1 = -12$

iv $S_{25} = 900$

14

a

i $a = -5$ ii $u_2 = 1$ iii $d = 6$

b $u_n = -11 + 6n$

15

a $d = -y - 4$

b $80 = y(y+d)(y+2d)$

c $y = -10$

d $-10, -4, 2$

Sigma notation

16

a $\sum_{k=1}^{5} \frac{1}{k^3}$

b $\sum_{k=1}^{7} \frac{1}{3k}$

c $\sum_{k=1}^{11} \frac{2}{5+k}$

17

a $\frac{47}{60}$ b $\frac{19}{20}$ c 15 d 14

18

a

i $2 + 10 + 30 + 68 + 130 + 222 + 350$

ii $1 + 2 + 3 + 4 + 5 + 6 + 7 + 1 + 8 + 27 + 64 + 125 + 216 + 343$

b Yes

19 No solution provided

Technology

20

a $S_{10} = 319.10$

b $S_{10} = 167.877$

c $S_{10} = -76.728$

21

a $u_1 = 57$

b $d = 14$

c $u_{34} = 519$

22

a $d = 3x + 2$

b $S_{14} = 301x + 224$

c $x = 2$

d $u_n = 7 + 8(n - 1)$

23

a

i $u_1 = 75$

ii $d = 24$

b

i $u_1 = 93$

ii $d = -25$

24

a $n = 26$

b $n = 22$

Applications

25 the sum is 180

26

a $3 + n$

b $\frac{n}{2}(7 + n)$ pipes

27

a 82 minutes

b $n = 12$

28

a $6k + 2$ metres

b $n(3n + 5)$ metres

c $n = 10$

29

a 4.5 km

b 112.25 km

30

a 430 kg

b 5390 kg

c 26